Clinton Gao

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Research interests: grounding & hallucination · tool-use & retrieval-augmented agents · evaluation methodology · LLM-as-judge reliability · failure analysis · NLP

EDUCATION

McGill University — B.Sc., Computer Science *Expected Winter 2027*

Montréal, QC · CGPA 3.71 / 4.00 · 30 advanced-standing credits (IB Diploma)

Relevant coursework: Natural Language Processing (A), Probability (A), Linear Algebra (A), Algorithms & Data Structures, Algorithm Design, Discrete Mathematics, Programming Languages & Paradigms, Operating Systems, Computer Systems.

COMP 400 Independent Research Project — planned, Fall 2026.

RESEARCH & PROJECTS

League AI Coach — LLM tool-using agent & trace-grounded evaluation *TypeScript*

Express · Zod · Supabase/Postgres · React · SSE · Anthropic / OpenAI / DeepSeek · github.com/xjdxx123/league-ai-coach

- Designed and built a 21-tool streaming LLM coaching agent over League of Legends match data, persisting every tool call, argument, and result per turn; currently extending toward a multi-agent match-analysis architecture (in progress).
- Engineered a trace-grounded evaluation harness combining deterministic checks, a hand-authored gold-match dataset, and an LLM-as-judge, with baseline / regression tracking across changes.
- Derived a 6-class failure taxonomy that guided successive mitigation rounds, improving the agent's grounding score from 46 to 83 (+37) on the gold set; a controlled ablation isolated pre-injection's causal effect as −31% tool calls, with grounding gains attributable to the always-on prompt rules rather than injection itself.
- Identified non-deterministic tool selection as the root cause of agent hallucination; mitigated via per-turn context injection as insurance against skip-then-fabricate.

Implicit Negation in Sentiment Classification — COMP 550 (NLP) research project *Python*

ACL-format paper · team of 3 (my role: model implementation, experiments, results) · BERT · PyTorch · HuggingFace

- Studied how traditional models (Naive Bayes, Logistic Regression) vs. fine-tuned BERT handle implicit negation — negative sentiment with no surface cue — on IMDb, with evaluation stratified by negation type (normal / explicit / implicit).
- Implemented and ran the five model configurations and a lexicon-augmentation experiment; the study found augmentation recovers ~72% of Logistic Regression's implicit-negation degradation but cannot close the gap to BERT.

Quantitative Factor-Investing System — research & backtesting tool (in progress) *Python*

- Built a factor-lab that computes asset returns and evaluates equity factors (e.g. 12-1 momentum), with a Probability-of-Backtest-Overfitting (PBO) guard against overfitting.

TECHNICAL SKILLS

Programming: Python, TypeScript, Java, JavaScript, SQL

AI / ML: LLM agents & tool-use, retrieval / grounding, LLM & model evaluation (LLM-as-judge), BERT / Transformers (HuggingFace), PyTorch, scikit-learn, prompt engineering, structured output (Zod)

Systems: Node.js, Express, Supabase / PostgreSQL, REST, SSE streaming, Docker, Git, CI/CD (GitHub Actions, Railway)

Quant / Data: factor models, backtesting, statistical evaluation, pandas / NumPy

Languages: Mandarin (native), English (fluent)